# FINAL EXPLANATION: MATHEMATICS GRADE 10

*Student Assessment Document*

| FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 2.3: AREA OF PART OF A CIRCLE | |
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| **Student Assessment Document** | |
| **Student Name** | ______________________________ |
| **Class** | ______________________________ |
| **Date** | ______________________________ |

| INSTRUCTIONS FOR STUDENTS |
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| This is your Final Explanation document. Your job is to write a well-structured, evidence-based response to the driving question below. Use everything you have learned across all 10 lessons — your predictions, observations, experiments, models, and class discussions — to construct a complete and convincing explanation. Write in full sentences and paragraphs. Support every claim you make with mathematical reasoning, correct formulas, and worked examples set in real Kenyan contexts. A strong Final Explanation does not just state facts — it explains WHY the mathematics works the way it does, and it shows HOW to apply it to solve real problems in farming, design, and everyday Kenyan life.<br><br>DRIVING QUESTION: How can we use the radius and the central angle of a circular pizza slice to calculate its exact area — and apply the same method to solve real problems in farming, design, and everyday life in Kenya?<br><br>INSTRUCTIONS FOR STUDENTS:<br>1. Answer each section in order. Each section builds on the previous one.<br>2. Use correct mathematical vocabulary throughout (sector, segment, arc, central angle, radius, proportion).<br>3. Show all working clearly — do not skip steps.<br>4. Use at least ONE real Kenyan context example in your explanation (e.g., a circular shamba in Kericho, a round chapati, a sprinkler system at a Rift Valley farm, a Maasai boma design).<br>5. Draw and label at least one diagram to support your explanation.<br>6. Connect your explanation back to the pizza/chapati phenomenon we started with — did your initial prediction turn out to be correct? What did the mathematics reveal? |

| Section 1: From the Pizza Slice to the Sector Formula — Explaining the Relationship | |
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| **Explain what a sector of a circle is and derive or explain the formula for its area. Start from what you already know about the area of a full circle. Why does the central angle control** | A sector is a 'slice' of a circle bounded by two radii and the arc between them — exactly like a slice of the circular chapati we studied at the start of this unit. To find its area, we start with what we know: the area of a full circle is $A = \pi r^2$, where r is the radius. A full circle sweeps a central angle of 360°. A sector sweeps only part of that full angle, so it covers only a fraction of the total area. That fraction is simply $(\theta \div 360°)$, where $\theta$ is the central angle of the sector. Putting this together gives the sector area formula: $A = (\theta/360°) \times \pi r^2$. For example, a sector with $\theta = 90°$ takes up 90/360 = 1/4 of the full circle, so its area is exactly $(1/4)\pi r^2$. This confirms the proportional relationship: because the formula is linear in $\theta$ ($\theta$ appears only once, multiplied by a constant), doubling $\theta$ |

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| **the fraction of the circle you are working with? Prove, using proportional reasoning, that a slice with twice the angle always has exactly twice the area — and explain why this proportional relationship holds perfectly. Reference the pizza/chapati phenomenon: when you first looked at the unequal slices, what did you predict, and what did the formula confirm or correct?** | always exactly doubles the area. If one chapati slice has $\theta = 60°$ and another has $\theta = 120°$, the second slice has exactly twice the area — every time, without exception. When we first looked at the unequal pizza slices, many of us guessed the relationship might not be perfectly proportional — perhaps the wider slice 'wastes' space near the crust. The formula shows this intuition was wrong: the relationship between angle and area is perfectly linear, because area scales uniformly as the sector 'opens up' around the fixed centre. |

## Section 2: Worked Calculation — Applying the Sector Formula to a Kenyan Context

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| **Choose ONE real Kenyan context (examples: a circular irrigation sprinkler covering part of a maize farm in the Rift Valley; a round chapati cut into unequal portions for a family in Nairobi; a sector-shaped plot of land on a tea estate in Kericho; a circular stage design for a school event). Clearly state the context, define the radius and central angle, then calculate the area of the sector step by step. Show all working. Round your final answer to 2 decimal places and include the correct units. Explain what your answer means in the context you chose.** | Context: A farmer in Kericho has installed a circular sprinkler at the centre of his tea farm. The sprinkler pipe reaches 14 metres in every direction (radius = 14 m). However, a fence blocks part of the farm, so the sprinkler can only rotate through an angle of 135°. What area of the tea farm does the sprinkler water? Step 1 — Write the formula: $A = (\theta/360°) \times \pi r^2$. Step 2 — Substitute known values: $A = (135/360) \times \pi \times 14^2$. Step 3 — Simplify the fraction: $135/360 = 3/8$. Step 4 — Calculate $r^2$: $14^2 = 196$. Step 5 — Multiply: $A = (3/8) \times \pi \times 196 = (3 \times 196\pi)/8 = 588\pi/8 = 73.5\pi$. Step 6 — Evaluate: $A = 73.5 \times 3.14159... \approx 230.91$ m². This means the sprinkler waters approximately 230.91 square metres of the tea farm. The farmer now knows exactly how much of his farm is being irrigated, which helps him decide whether he needs a second sprinkler to cover the remaining area — a very practical use of the sector formula. |

## Section 3: Area of a Segment — Going Beyond the Sector

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| **Explain the difference between a sector and a segment of a circle. Derive or explain the formula for the area of a minor segment. Why must you subtract the area of a triangle** | A sector is the full 'pie slice' shape — bounded by two radii and an arc. A segment is the region between a chord and the arc it cuts off — like the rounded 'edge piece' left over when you cut straight across a circular chapati rather than cutting through the centre. To find the area of a minor segment, you take the area of the sector that contains it and subtract the area of the triangle formed by the two radii and the chord: Area of segment = Area of sector − Area of triangle = $(\theta/360°)\pi r^2 - (1/2)r^2 \sin\theta$. The subtraction is necessary because the sector includes the triangular region between the two radii, but the segment does not — it is only the curved 'cap' above the chord. [DIAGRAM: Circle with centre O, two radii OA and OB, chord AB, sector OAB shaded, triangle OAB |

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| **from the sector area? Show a clearly labelled diagram. Then solve a worked example: a circular water tank in Kisumu has a radius of 5 m; a straight pipe cuts across the tank, creating a chord that subtends a central angle of 80°. Calculate the area of the smaller segment of water cut off by the pipe. Show all steps.** | labelled, segment (region between chord AB and arc AB) cross-hatched.] Worked example — Kisumu water tank: $r = 5$ m, $\theta = 80°$. Sector area = $(80/360) \times \pi \times 5^2 = (2/9) \times 25\pi = 50\pi/9 \approx 17.45$ m². Triangle area = $(1/2) \times r^2 \times \sin\theta = (1/2) \times 25 \times \sin 80° = 12.5 \times 0.9848 \approx 12.31$ m². Segment area = $17.45 - 12.31 \approx 5.14$ m². This tells the Kisumu water engineer that approximately 5.14 m² of the tank's cross-section lies on the smaller side of the pipe — useful for calculating the volume of water in that section. |

| Section 4: The Proportional Relationship — Deeper Mathematical Reasoning | |
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| **Return to the central puzzle from our phenomenon: 'Is a slice with twice the angle always exactly twice the area?' Answer this question with full mathematical justification — not just by example, but by using the formula algebraically. Then extend your thinking: does the same proportional relationship hold between the arc length and the central angle? Show the arc length formula $L = (\theta/360°) \times 2\pi r$ and confirm whether arc length and sector area scale in the same way with respect to θ. Finally, explain what would happen to the sector area if you kept θ constant but doubled the radius — is the relationship between radius and area also proportional? What does this mean practically for a farmer or designer in Kenya?** | To prove algebraically that doubling the angle exactly doubles the area: let a sector have central angle θ and radius r. Its area is $A_1 = (\theta/360°) \times \pi r^2$. Now consider a second sector with the same radius but angle 2θ. Its area is $A_2 = (2\theta/360°) \times \pi r^2 = 2 \times (\theta/360°) \times \pi r^2 = 2A_1$. This confirms perfectly that $A_2 = 2A_1$ — doubling θ always exactly doubles the area. This works because θ appears as a linear factor in the formula. Arc length follows the same pattern: $L = (\theta/360°) \times 2\pi r$. Doubling θ gives $L_2 = (2\theta/360°) \times 2\pi r = 2L_1$. So yes, both sector area and arc length scale proportionally and linearly with θ. However, the relationship between radius and area is different. If we keep θ fixed and double the radius from r to 2r: $A\_new = (\theta/360°) \times \pi(2r)^2 = (\theta/360°) \times 4\pi r^2 = 4A_1$. Doubling the radius quadruples the area — the relationship is quadratic ($r^2$), not linear. This has a very important practical implication: a farmer in the Rift Valley who extends her sprinkler pipe from 10 m to 20 m does not just double her irrigated area — she quadruples it. Understanding this difference between linear scaling (angle) and quadratic scaling (radius) is essential for efficient farm and design planning in Kenya. |

| Section 5: Connecting Everything — Answering the Driving Question | |
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| **Now bring everything together. Write a complete, connected** | The driving question asks how we can use the radius and central angle of a circular slice to calculate its exact area, and how we can apply this to real life in Kenya. The answer lies in the sector area formula: $A = (\theta/360°) \times \pi r^2$, where r is the radius of the full |

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| **answer to the driving question. Your response must: (a) state the sector area formula and explain every part of it; (b) explain how the same formula applies across different real Kenyan contexts (farming, food, design — give at least two specific examples); (c) reflect on your learning journey — what did you predict at the start of the unit when you saw the unequal chapati/pizza slices, what confused you, and what did the mathematics ultimately reveal? (d) state one limitation of the sector/segment model in real life (e.g., a pizza slice is never a perfect geometric sector) and explain how a mathematician or engineer would handle this. End with a strong concluding statement.** | circle and θ is the central angle of the sector in degrees. Every part of this formula matters: $\pi r^2$ gives the area of the complete circle; the fraction θ/360° tells us exactly what proportion of that full circle our sector represents. Together, they give a precise, reliable calculation for any sector, no matter the size. This formula applies across many Kenyan contexts. For a tea farmer in Kericho installing a rotating sprinkler, the formula tells her exactly how many square metres her equipment will irrigate — allowing her to budget for water and fertiliser accurately. For a jua kali artisan in Nairobi designing a circular metal tray with decorative sector-shaped panels, the formula ensures each panel has the correct area so the design is balanced and materials are not wasted. For a chef in Mombasa cutting a large round mkate (bread) into portions of specific sizes, the formula guarantees fair, equal servings by angle. At the start of this unit, when we saw the unequal chapati slices on the table, many of us guessed that the wider slices were 'more than proportionally' larger — that the relationship might curve or vary. The formula corrected this: the relationship between angle and area is perfectly linear, clean, and proportional. What did confuse us initially was the difference between sectors and segments, and why you must subtract a triangle to find a segment's area. Working through examples step by step made this clear. One real limitation is that physical objects — a pizza, a chapati, a farm plot — are never perfect geometric circles or sectors. A pizza slice may have an uneven crust, a farm boundary may curve irregularly, a sprinkler may not rotate with a perfectly fixed radius. In these cases, mathematicians and engineers use approximation methods, breaking irregular shapes into smaller sectors or using integration in advanced mathematics. Nevertheless, the sector formula remains the essential starting point and gives an excellent approximation for any problem involving a roughly circular shape. In summary, the radius and the central angle together fully determine the area of any sector — and this simple, powerful relationship, expressed as $A = (\theta/360°) \times \pi r^2$, is a tool that connects the mathematics classroom directly to farming, design, and everyday life across Kenya. |

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| **Criterion** | **Excellent (4)** | **Proficient (3)** | **Developing (1–2)** |
| Mathematical Accuracy — Formulas and Calculations | Sector area formula ($A = (\theta/360°)\pi r^2$) and segment formula (sector area − triangle area) are stated and applied correctly in all sections. All calculations are accurate, fully worked, and rounded correctly with appropriate units. No mathematical errors. | Both formulas are correctly stated and applied in most sections. Minor arithmetic errors may be present but do not affect overall reasoning. Units are mostly correct. | Formulas are partially recalled or misapplied. Calculations contain significant errors or missing steps. Units are missing or incorrect in multiple places. |
| Depth of Mathematical Reasoning and Justification | The student explains WHY the proportional relationship between angle and area is linear, using algebraic proof (not just examples). The quadratic relationship between radius and area is correctly identified and explained. All claims are fully justified. | The student demonstrates understanding of proportionality using examples and some algebraic reasoning. The distinction between linear (angle) and quadratic (radius) scaling is mentioned but may not be fully justified. | The student states the proportional relationship without justification, or justifies it only with a single numeric example. The difference between how angle and radius affect area is not clearly addressed. |
| Application to Real Kenyan Contexts | At least two distinct, specific, and realistic Kenyan contexts are used (e.g., Kericho | At least one Kenyan context is used with a complete worked example. A second | A Kenyan context is named but not meaningfully connected to the |

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| | farm, Nairobi artisan, Mombasa food). Contexts are fully integrated into calculations — not just mentioned as decoration. The explanation of what the mathematical result means in context is clear and insightful. | context is mentioned but may not be fully developed. The practical meaning of the result is stated. | mathematics. The example is generic or the context is used only superficially without explaining the real-world significance. |
| Use of Diagrams and Mathematical Vocabulary | At least one clearly drawn and fully labelled diagram is included (showing centre, radii, arc, chord, sector, and/or segment as appropriate). Correct mathematical vocabulary is used consistently throughout: sector, segment, arc, central angle, radius, proportion, chord. | A diagram is included and mostly labelled correctly, though one or two labels may be missing. Most key vocabulary terms are used correctly. Minor inconsistencies in terminology. | A diagram is missing, unlabelled, or incorrectly drawn. Mathematical vocabulary is limited, inconsistent, or used incorrectly in several places. |
| Reflection and Connection to the Driving Question | The final section directly and fully answers the driving question. The student clearly reflects on their initial prediction from the pizza/chapati phenomenon, identifies what the mathematics confirmed or corrected, and acknowledges at least one real-world limitation of the model. The concluding statement is strong and specific. | The driving question is addressed in the final section. Some reflection on initial predictions is included. A limitation is mentioned. The conclusion is present but may be general rather than specific. | The final section restates facts from earlier sections without directly answering the driving question. Reflection on the phenomenon or initial predictions is absent or very brief. No limitation is identified. |